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Kelas : IF-1

LISTING PRAKTIKUM STRUKTUR DATA
MODUL 10
DOUBLE LINKEDLIST (SENARAI BERANTAI GANDA)

PRAKTIK

Praktik 1

```
import java.util.Scanner;

class simpul {
    String nama;
    String alamat;
    int umur;
    char jekel;
    String hobi[] = new String[3];
    float ipk;
    simpul kiri;
    simpul kanan;
}

class senaraiGanda {
    public static simpul awal;
    public static simpul akhir;

    public static void inisialisasiSenaraiKosong() {
        awal = null;
        akhir = null;
    }

    public static void tambahDepan() {
        Scanner masukan = new Scanner(System.in);
        int bacaTombol = 0;

        System.out.println("TAMBAH DEPAN : ");
        System.out.print("Silakan masukkan nama anda : ");
        String NAMA = masukan.nextLine();
        System.out.print("Silakan masukkan alamat anda : ");
        String ALAMAT = masukan.nextLine();
        System.out.print("Silakan masukkan umur anda : ");
        int UMUR = masukan.nextInt();
        System.out.print("Silakan masukkan jenis Kelamin anda : ");
        try {
            bacaTombol = System.in.read();
        } catch (java.io.IOException e) {
        }
        char JEKEL = (char) bacaTombol;
        masukan.nextLine(); // membersihkan newline

        System.out.println("Silakan masukkan hobi (maks 3) : ");
```

```

System.out.print("hobi ke-0 : ");
String HOBI0 = masukan.next();
System.out.print("hobi ke-1 : ");
String HOBI1 = masukan.next();
System.out.print("hobi ke-2 : ");
String HOBI2 = masukan.next();
System.out.print("Silakan masukkan IPK anda : ");
float IPK = masukan.nextFloat();

simpul baru = new simpul();
baru.nama = NAMA;
baru.alamat = ALAMAT;
baru.umur = UMUR;
baru.jekel = JEKEL;
baru.hobi[0] = HOBI0;
baru.hobi[1] = HOBI1;
baru.hobi[2] = HOBI2;
baru.ipk = IPK;

if (awal == null) {
    awal = baru;
    akhir = baru;
    baru.kiri = null;
    baru.kanan = null;
} else {
    baru.kanan = awal;
    awal.kiri = baru;
    awal = baru;
    awal.kiri = null;
}
}

public static void tambahBelakang() {
    Scanner masukan = new Scanner(System.in);
    int bacaTombol = 0;

    System.out.println("TAMBAH BELAKANG : ");
    System.out.print("Silakan masukkan nama anda : ");
    String NAMA = masukan.nextLine();
    System.out.print("Silakan masukkan alamat anda : ");
    String ALAMAT = masukan.nextLine();
    System.out.print("Silakan masukkan umur anda : ");
    int UMUR = masukan.nextInt();
    System.out.print("Silakan masukkan Jenis Kelamin anda : ");
    try {
        bacaTombol = System.in.read();
    } catch (java.io.IOException e) {
    }
    char JEKEL = (char) bacaTombol;
    masukan.nextLine();

    System.out.println("Silakan masukkan hobi (maks 3) : ");
    System.out.print("hobi ke-0 : ");
    String HOBI0 = masukan.next();
    System.out.print("hobi ke-1 : ");
    String HOBI1 = masukan.next();
    System.out.print("hobi ke-2 : ");
    String HOBI2 = masukan.next();
    System.out.print("Silakan masukkan IPK anda : ");
    float IPK = masukan.nextFloat();

    simpul baru = new simpul();
    baru.nama = NAMA;
    baru.alamat = ALAMAT;
    baru.umur = UMUR;
    baru.jekel = JEKEL;
    baru.hobi[0] = HOBI0;
    baru.hobi[1] = HOBI1;
    baru.hobi[2] = HOBI2;
    baru.ipk = IPK;

```

```

    if (awal == null) {
        awal = baru;
        akhir = baru;
        baru.kiri = null;
        baru.kanan = null;
    } else {
        baru.kiri = akhir;
        akhir.kanan = baru;
        akhir = baru;
        akhir.kanan = null;
    }
}

public static int hitungJumlahSimpul() {
    int N = 0;
    simpul bantu = awal;
    while (bantu != null) {
        N++;
        bantu = bantu.kanan;
    }
    return N;
}

public static void tambahTengah() {
    Scanner masukan = new Scanner(System.in);
    System.out.println("Tentukan Lokasi Penambahan Data");
    int LOKASI = masukan.nextInt();
    masukan.nextLine();

    int jumlahSimpulYangAda = hitungJumlahSimpul();
    if (LOKASI == 1) {
        System.out.println("Lakukan penambahan di depan");
    } else if (LOKASI > jumlahSimpulYangAda) {
        System.out.println("Lakukan penambahan di belakang");
    } else {
        System.out.println("TAMBAH TENGAH : ");
        System.out.print("Silakan masukkan nama anda : ");
        String NAMA = masukan.nextLine();
        System.out.print("Silakan masukkan alamat anda : ");
        String ALAMAT = masukan.nextLine();
        System.out.print("Silakan masukkan umur anda : ");
        int UMUR = masukan.nextInt();
        System.out.print("Silakan masukkan Jenis Kelamin anda : ");
        int bacaTombol = 0;
        try {
            bacaTombol = System.in.read();
        } catch (java.io.IOException e) {
        }
        char JEKEL = (char) bacaTombol;
        masukan.nextLine();

        System.out.println("Silakan masukkan hobi (maks 3) : ");
        System.out.print("hobi ke-0 : ");
        String HOBI0 = masukan.next();
        System.out.print("hobi ke-1 : ");
        String HOBI1 = masukan.next();
        System.out.print("hobi ke-2 : ");
        String HOBI2 = masukan.next();
        System.out.print("Silakan masukkan IPK anda : ");
        float IPK = masukan.nextFloat();

        simpul bantu = awal;
        int i = 1;
        while (i < LOKASI && bantu != akhir) {
            bantu = bantu.kanan;
            i++;
        }

        simpul baru = new simpul();

```

```

        baru.nama = NAMA;
        baru.alamat = ALAMAT;
        baru.umur = UMUR;
        baru.jekel = JEKEL;
        baru.hobi[0] = HOBI0;
        baru.hobi[1] = HOBI1;
        baru.hobi[2] = HOBI2;
        baru.ipk = IPK;

        baru.kiri = bantu.kiri;
        baru.kiri.kanan = baru;
        baru.kanan = bantu;
        bantu.kiri = baru;
    }
}

public static void hapus() {
    if (awal == null) // jika senarai masih kosong
    {
        System.out.println("senarai kosong, menghapus tidak dapat dilakukan");
    } else // jika senarai tidak kosong
    {

        Scanner masukan = new Scanner(System.in);
        System.out.print("Silakan masukkan nama yang ingin dihapus : ");
        String NAMACARI = masukan.nextLine();

        if (awal == akhir) // jika hanya ada sebuah simpul
        {
            if (awal.nama.equals(NAMACARI)) {
                System.out.println("menghapus " + NAMACARI + " dilakukan..");
                inisialisasiSenaraiKosong();
            } else
                System.out.println("data " + NAMACARI + " tidak ditemukan");
        }

        else if (awal.nama.equals(NAMACARI)) // jika nama ditemukan di awal
        {
            System.out.println("menghapus " + NAMACARI + " dilakukan..");
            awal = awal.kanan;
        } else {
            simpul bantu;
            bantu = awal;
            while (bantu.kanan.nama.equals(NAMACARI) == false) {
                bantu = bantu.kanan;
                if (bantu.kanan == null)
                    break;
            }
            if ((bantu == akhir) && (akhir.nama.equals(NAMACARI) == false)) {
                System.out.println("data " + NAMACARI + " tidak ditemukan");
            } else if (akhir.nama.equals(NAMACARI)) // jika nama ditemukan di akhir
            {
                bantu.kanan = null;
                akhir = bantu;
            } else {
                System.out.println("menghapus " + NAMACARI + " dilakukan..");
                bantu.kanan = bantu.kanan.kanan;
            }
        }
    }
}

public static void cetakSenaraiMaju() {
    if (awal == null) {
        System.out.println("...MAAF SENARAI KOSONG....");
    } else {
        System.out.println("NO NAMA ALAMAT UMUR JEKEL HOBI[0] HOBI[1] HOBI[2] IPK");
        simpul bantu = awal;
        while (bantu != null) {
            System.out.print(bantu.nama + "\t");

```

```

        System.out.print(bantu.alamat + "\t");
        System.out.print(bantu.umur + "\t");
        System.out.print(bantu.jekel + "\t");
        System.out.print(bantu.hobi[0] + "\t");
        System.out.print(bantu.hobi[1] + "\t");
        System.out.print(bantu.hobi[2] + "\t");
        System.out.println(bantu.ipk);
        bantu = bantu.kanan;
    }
}

public static void cetakSenaraiMundur() {
    if (awal == null) {
        System.out.println("...MAAF SENARAI KOSONG....");
    } else {
        System.out.println("NO NAMA ALAMAT UMUR JEKEL HOBI[0] HOBI[1] HOBI[2] IPK");
        simpul bantu = akhir;
        while (bantu != null) {
            System.out.print(bantu.nama + "\t");
            System.out.print(bantu.alamat + "\t");
            System.out.print(bantu.umur + "\t");
            System.out.print(bantu.jekel + "\t");
            System.out.print(bantu.hobi[0] + "\t");
            System.out.print(bantu.hobi[1] + "\t");
            System.out.print(bantu.hobi[2] + "\t");
            System.out.println(bantu.ipk);
            bantu = bantu.kiri;
        }
    }
}

public static void main(String[] args) {
    inisialisasiSenaraiKosong();
    tambahDepan();
    tambahDepan();
    tambahBelakang();
    tambahBelakang();
    tambahTengah();
    System.out.println("HAPUS DATA");
    hapus();
    System.out.println(" ");
    System.out.println("CETAK MAJU");
    cetakSenaraiMaju();
    System.out.println(" ");
    System.out.println("CETAK MUNDUR");
    cetakSenaraiMundur();
}
}

```

Run | Debug

```
public static void main(String[] args) {  
    inisialisasiSenaraiKosong();  
  
    tambahDepan();  
    tambahDepan();  
    tambahBelakang();  
    tambahBelakang();  
    tambahTengah();  
    System.out.println(x: "HAPUS DATA");  
    hapus();  
    cetakSenaraiMaju();  
}
```

Tentukan Lokasi Penambahan Data

2

TAMBAH TENGAH :

Silakan masukkan nama anda : e

Silakan masukkan alamat anda : e

Silakan masukkan umur anda : 4

Silakan masukkan Jenis Kelamin anda : e

Silakan masukkan hobi (maks 3) :

hobi ke-0 : e

hobi ke-1 : e

hobi ke-2 : e

Silakan masukkan IPK anda : 4

HAPUS DATA

Silakan masukkan nama yang ingin dihapus : d

NO NAMA ALAMAT UMUR JEKEL HOBI[0] HOBI[1] HOBI[2] IPK

b	b	4	b	b	b	b	4.0
---	---	---	---	---	---	---	-----

e	e	4	e	e	e	e	4.0
---	---	---	---	---	---	---	-----

a	a	4	a	a	a	a	4.0
---	---	---	---	---	---	---	-----

c	c	4	c	c	cc	c	5.0
---	---	---	---	---	----	---	-----

PS D:\AREAS\Semester3\Struktur data\10>

eady

Praktik 2

Run | Debug

```
public static void main(String[] args) {  
    inisialisasiSenaraiKosong();  
  
    tambahDepan();  
    tambahDepan();  
    tambahBelakang();  
    tambahBelakang();  
    tambahTengah();  
    System.out.println(x: "HAPUS DATA");  
    hapus();  
    System.out.println(x: " ");  
    System.out.println(x: "CETAK MAJU");  
    cetakSenaraiMaju();  
    System.out.println(x: " ");  
    System.out.println(x: "CETAK MUNDUR");  
    cetakSenaraiMundur();  
}
```

Tentukan Lokasi Penambahan Data

2

TAMBAH TENGAH :

Silakan masukkan nama anda : e

Silakan masukkan nama anda : e

Silakan masukkan alamat anda : e

Silakan masukkan umur anda : 3

Silakan masukkan Jenis Kelamin anda : e

Silakan masukkan hobi (maks 3) :

hobi ke-0 : e

hobi ke-1 : e

hobi ke-2 : e

Silakan masukkan IPK anda : 4

HAPUS DATA

Silakan masukkan nama yang ingin dihapus : c

menghapus c dilakukan..

CETAK MAJU

NO	NAMA	ALAMAT	UMUR	JEKEL	HOBI[0]	HOBI[1]	HOBI[2]	IPK
----	------	--------	------	-------	---------	---------	---------	-----

b	b		4	b	b	b	b	4.0
---	---	--	---	---	---	---	---	-----

e	e		3	e	e	e	e	4.0
---	---	--	---	---	---	---	---	-----

a	a		4	a	a	aa	a	5.0
---	---	--	---	---	---	----	---	-----

dd	d		4	d	d	d	d	4.0
----	---	--	---	---	---	---	---	-----

CETAK MUNDUR

NO	NAMA	ALAMAT	UMUR	JEKEL	HOBI[0]	HOBI[1]	HOBI[2]	IPK
----	------	--------	------	-------	---------	---------	---------	-----

dd	d		4	d	d	d	d	4.0
----	---	--	---	---	---	---	---	-----

c	c		4	c	c	c	c	4.0
---	---	--	---	---	---	---	---	-----

a	a		4	a	a	aa	a	5.0
---	---	--	---	---	---	----	---	-----

e	e		3	e	e	e	e	4.0
---	---	--	---	---	---	---	---	-----

b	b		4	b	b	b	b	4.0
---	---	--	---	---	---	---	---	-----

PS D:\AREAS\Semester3\Struktur data\10>

Praktik 3

Praktik 4

Praktik 5

LATIHAN

Latihan 1

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TUGAS

Tugas 1

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